

```
In [1]: import pandas as pd
import numpy as np
```

```
In [2]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: df = pd.read_csv("Files/museum_visitors.csv", index_col=0)
df.head()
```

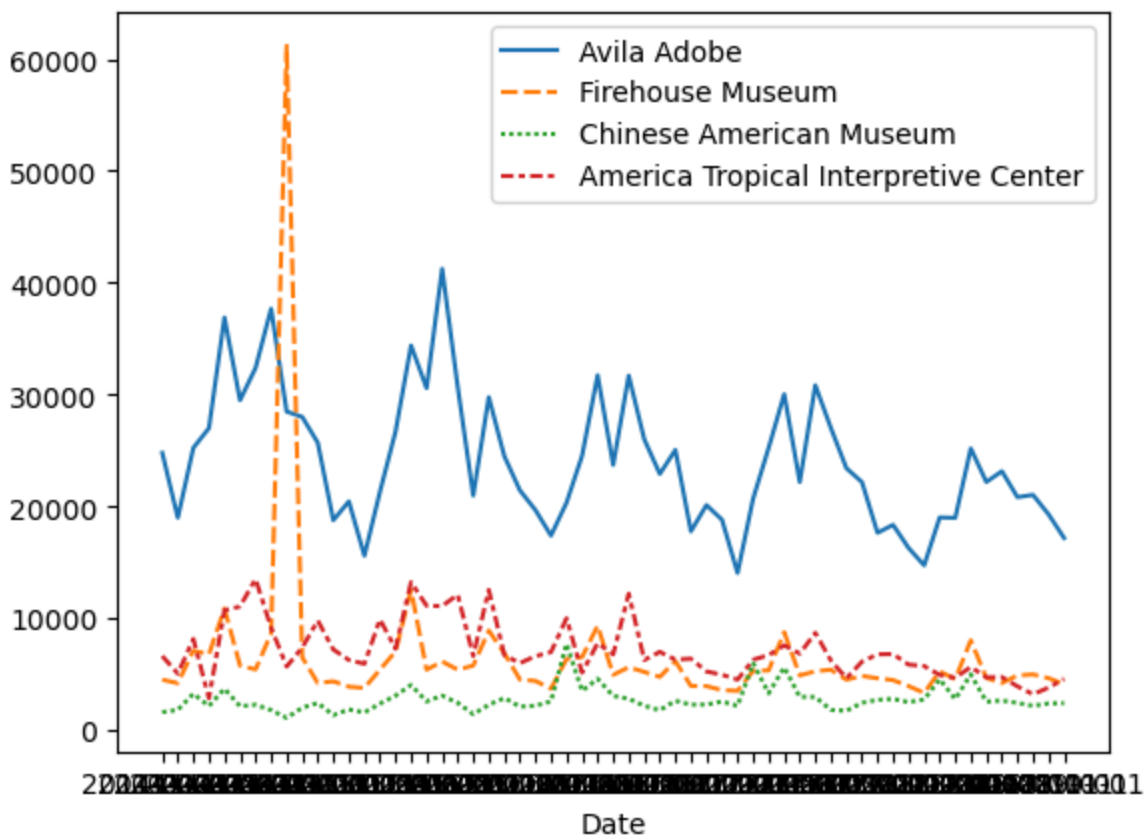
```
Out[3]:
```

	Avila Adobe	Firehouse Museum	Chinese American Museum	America Tropical Interpretive Center
Date				
2014-01-01	24778	4486	1581	6602
2014-02-01	18976	4172	1785	5029
2014-03-01	25231	7082	3229	8129
2014-04-01	26989	6756	2129	2824
2014-05-01	36883	10858	3676	10694

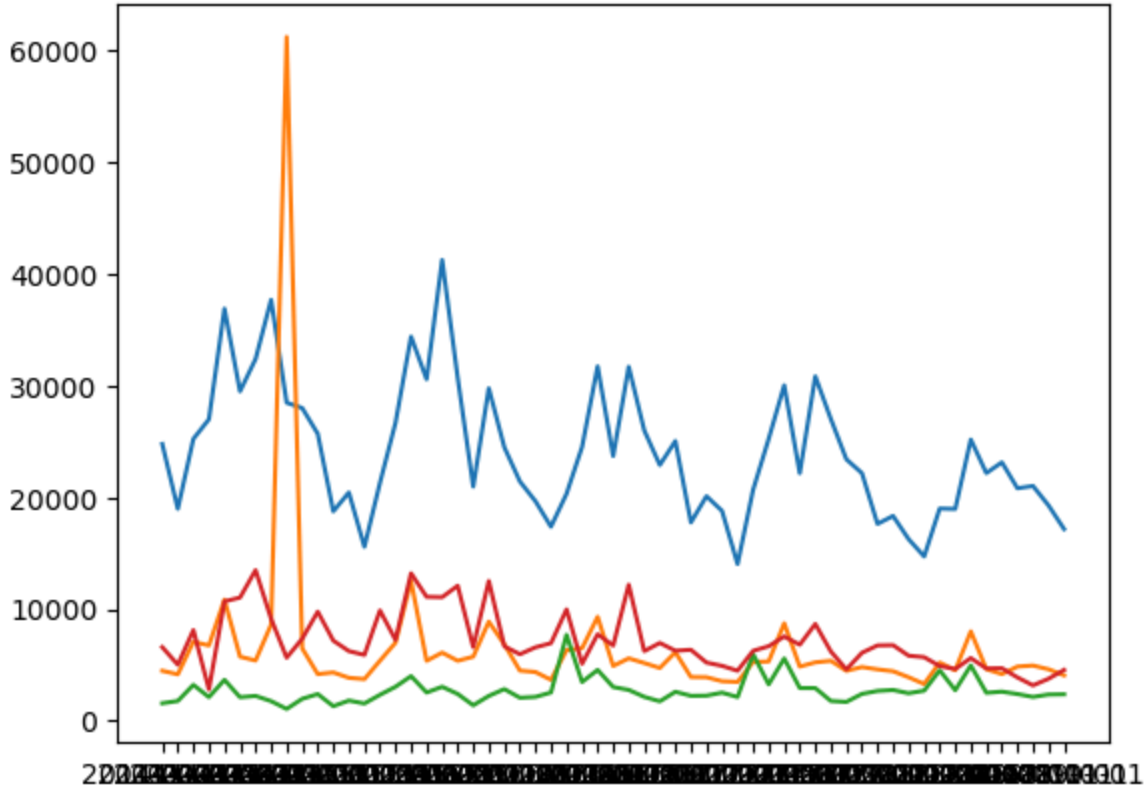
```
In [4]: df.shape
```

```
Out[4]: (59, 4)
```

```
In [16]: sns.lineplot(data = df) # sns.Lineplot(df) ham ishlaydi
plt.show()
```

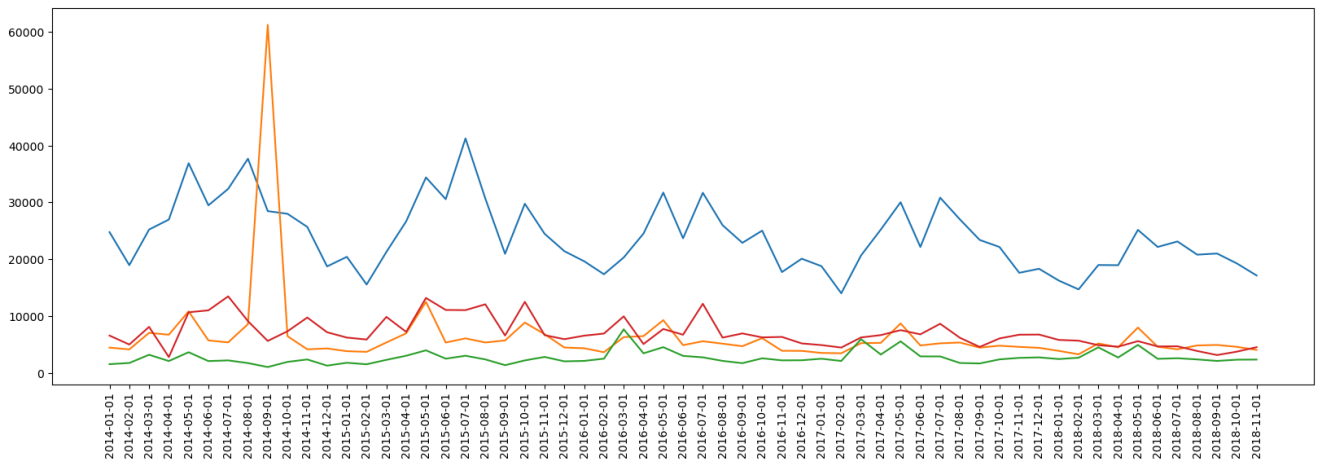


```
In [18]: plt.plot(df)
plt.show()
```



```
In [19]: plt.figure(figsize = (20, 6)) # plt.figure((16, 6)) - xatolik qaytaradi
plt.plot(df)
plt.xticks(rotation = 90)
```

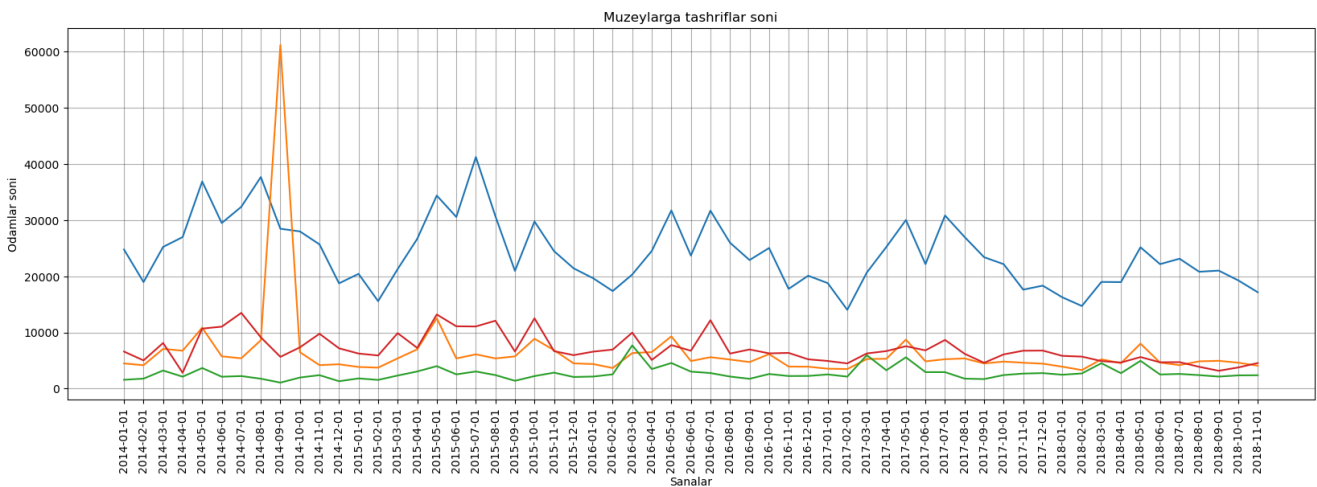
```
plt.show()
```



```
In [24]: plt.figure(figsize = (20, 6)) # plt.figure((16, 6)) - xatolik qaytaradi
```

```
plt.title("Muzeylarga tashriflar soni")
plt.xlabel("Sanalar")
plt.ylabel("Odamlar soni")
plt.grid(alpha = 0.3, color = "k")
plt.plot(df)
plt.xticks(rotation = 90)

plt.show()
```

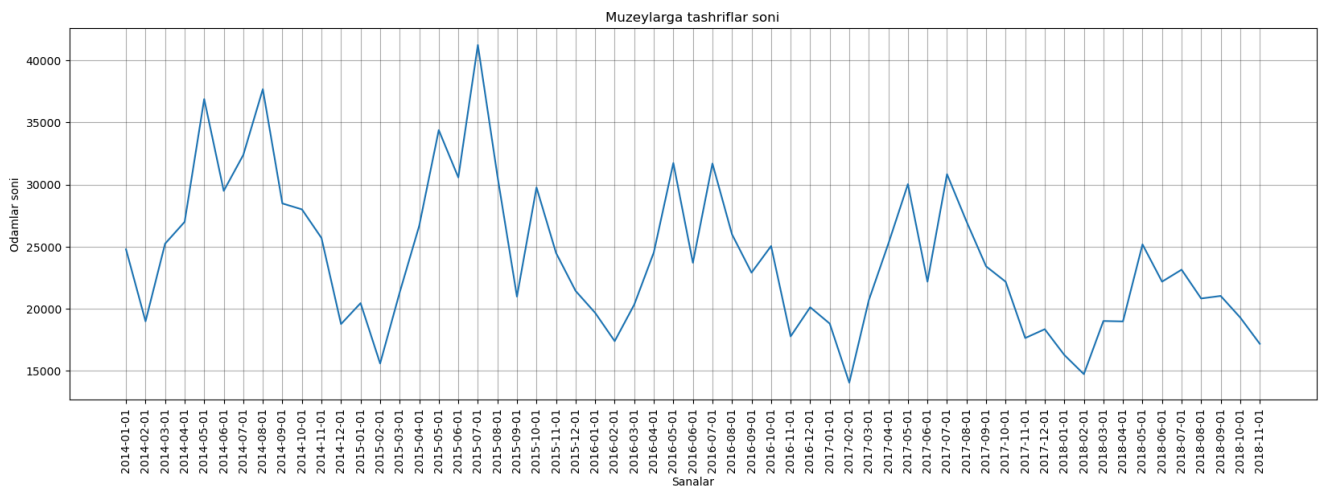


```
In [29]: plt.figure(figsize = (20, 6)) # plt.figure((16, 6)) - xatolik qaytaradi
```

```
plt.title("Muzeylarga tashriflar soni")
plt.xlabel("Sanalar")
plt.ylabel("Odamlar soni")
plt.grid(alpha = 0.3, color = "k")

sns.lineplot(data=df['Avila Adobe'])
plt.xticks(rotation = 90)

plt.show()
```



1. Ustunni indeks (raqam) bo'yicha chaqirish – `df.iloc` orqali

Agar siz 0-ustun yoki 2-ustunni chaqirmoqchi bo'lsangiz:

```
df.iloc[:, [0, 2]]
```

Buni Seaborn'da ishlatish:

```
sns.lineplot(data=df.iloc[:, [0, 2]])
```

`:` → barcha qatorlar `[0, 2]` → 0-ustun va 2-ustun

2. Faqat bitta ustun kerak bo'lsa

```
sns.lineplot(data=df.iloc[:, 0]) # 0-ustun
```

```
sns.lineplot(data=df.iloc[:, 2]) # 2-ustun
```

■ Qo'shimcha: Agar ustunlarni nom bilan chaqirmoqchi bo'lsangiz

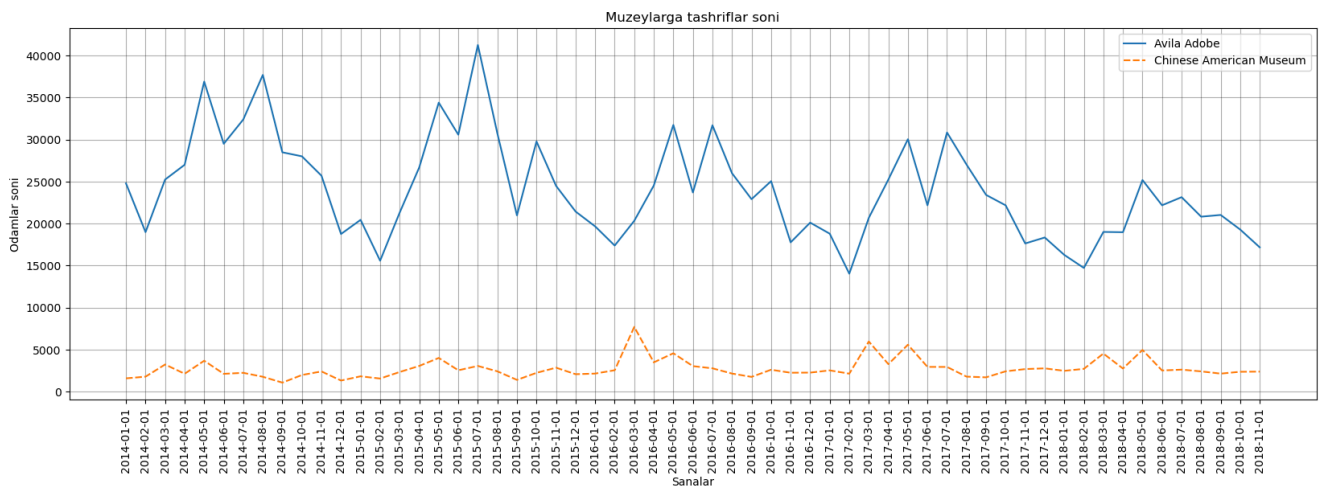
```
sns.lineplot(data=df[['col1', 'col3']])
```

In [31]: `plt.figure(figsize = (20, 6)) # plt.figure((16, 6)) - xatolik qaytaradi`

```
plt.title("Muzeylarga tashriflar soni")
plt.xlabel("Sanalar")
plt.ylabel("Odamlar soni")
plt.grid(alpha = 0.3, color = "k")
```

```
sns.lineplot(data=df.iloc[:, [0, 2]])
plt.xticks(rotation = 90)
```

```
plt.show()
```



loc nima?

`loc` — Pandas DataFrame'da **qator va ustunlarni NOMI bo'yicha** tanlash uchun ishlatiladi.

Sintaksis:

```
df.loc[ qatorlar , ustunlar ]
```

- Qatorlar → index nomi (yoki shart)
- Ustunlar → column nomi

QATORLARNI tanlash (`loc` bilan)

1. Bitta qatorni nomi bo'yicha olish

```
df.loc['row1']
```

2. Bir nechta qator

```
df.loc[['row1', 'row3', 'row5']]
```

3. Qatorlar bo'yicha interval

```
df.loc['row2' : 'row6']
```

Diapazon **oxirigacha kiradi** (INTEGER `iloc`dan farqli ravishda).

2 USTUNLARNI tanlash (loc bilan)

✓ 1. Bitta ustun

```
df.loc[:, 'Age']
```

✓ 2. Bir nechta ustun

```
df.loc[:, ['Age', 'Salary']]
```

✓ 3. Ustunlar diapazoni

```
df.loc[:, 'Age' : 'Salary']
```

3 Qator + ustun tanlash

✓ 1. Bitta katakni olish

```
df.loc['row1', 'Age']
```

✓ 2. Bir nechta qator + bir nechta ustun

```
df.loc[['row1', 'row3'], ['Age', 'Salary']]
```

4 Shartlar bilan loc ishlatish

✓ 1. Filtr: Age > 30

```
df.loc[df['Age'] > 30]
```

✓ 2. Filtr + ustun tanlash

```
df.loc[df['Age'] > 30, ['Name', 'Age']]
```

✓ 3. Ko'p shartlar

```
df.loc[(df['Age'] > 30) & (df['Salary'] < 5000)]
```

5 Qiymatlarni o'zgartirish (loc orqali)

✓ 1. Bitta katakni o'zgartirish

```
df.loc['row1', 'Age'] = 99
```

✓ 2. Shart bo'yicha ustun qiymatini o'zgartirish

```
df.loc[df['Age'] < 18, 'Status'] = 'Minor'
```

6 loc va iloc farqi

Xususiyat	loc	iloc
Indekslash turi	nom bo'yicha	raqam bo'yicha
Qator diapazoni	yakunini ham oladi	yakunini olmaydi
Sintaksis	<code>df.loc['A':'C']</code>	<code>df.iloc[0:2]</code>

7 Mini DataFrame bilan misollar

```
import pandas as pd
```

```
df = pd.DataFrame({  
    "Age": [22, 35, 40],  
    "Salary": [3000, 5000, 7000]  
}, index=["Ali", "Vali", "Sodiq"])
```

✓ Ali'ning yoshini olish:

```
df.loc['Ali', 'Age']
```

✓ Ali va Vali'ning maoshlarini olish:

```
df.loc[['Ali', 'Vali'], ['Salary']]
```

✓ Salary > 4000 bo'lganlarni olish:

```
df.loc[df['Salary'] > 4000]
```

Seabornda BARPLOT

```
In [38]: df = pd.read_csv("Files/uzbekistan.csv", index_col = 0)
df
```

```
Out[38]:
```

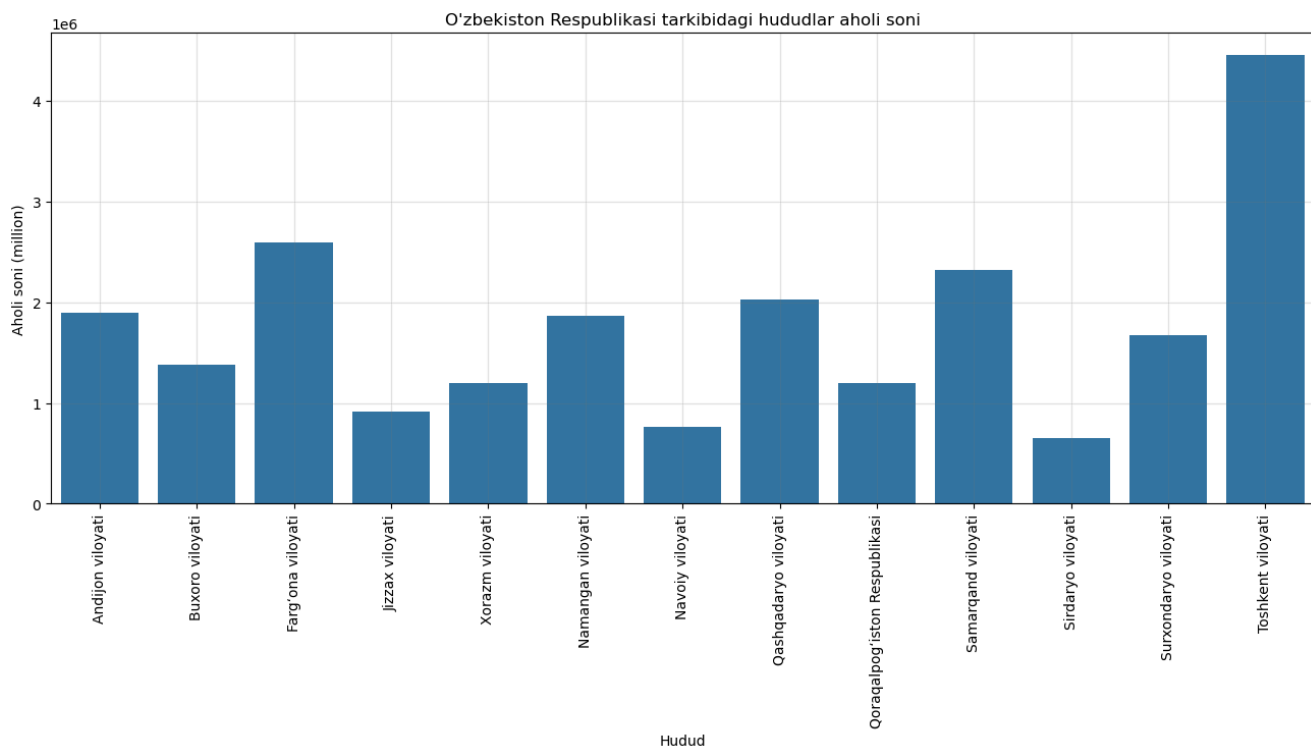
	Maydoni (kv.km)	Aholisi
Bo'linishi		
Andijon viloyati	4200.0	1899000.0
Buxoro viloyati	39400.0	1384700.0
Farg'ona viloyati	6800.0	2597000.0
Jizzax viloyati	20500.0	910500.0
Xorazm viloyati	6300.0	1200000.0
Namangan viloyati	7900.0	1862000.0
Navoiy viloyati	110800.0	767500.0
Qashqadaryo viloyati	28400.0	2029000.0
Qoraqalpog'iston Respublikasi	160000.0	1200000.0
Samarqand viloyati	16400.0	2322000.0
Sirdaryo viloyati	5100.0	648100.0
Surxondaryo viloyati	20800.0	1676000.0
Toshkent viloyati	15300.0	4450000.0

`index_col` — bu `pd.read_csv()` funksiyasida **CSV fayldan qaysi ustunni DataFrame indeksiga aylantirish kerakligini** belgilaydi.

Parametr	Ma'nosi
<code>index_col=0</code>	CSV dagi birinchi ustunni DataFrame indeksiga aylantiradi
<code>index_col=1</code>	ikkinchi ustun indeks bo'ladi
<code>index_col="Region"</code>	ustunni nomi orqali indeksga aylantirish

```
In [51]: plt.figure(figsize = (16, 6))
sns.barplot(x = df.index, y = df['Aholisi'])

plt.title("O'zbekiston Respublikasi tarkibidagi hududlar aholi soni")
plt.xlabel("Hudud")
plt.ylabel("Aholi soni (million)")
plt.grid(alpha = 0.2, color = "grey")
plt.xticks(rotation = 90)
plt.show()
```

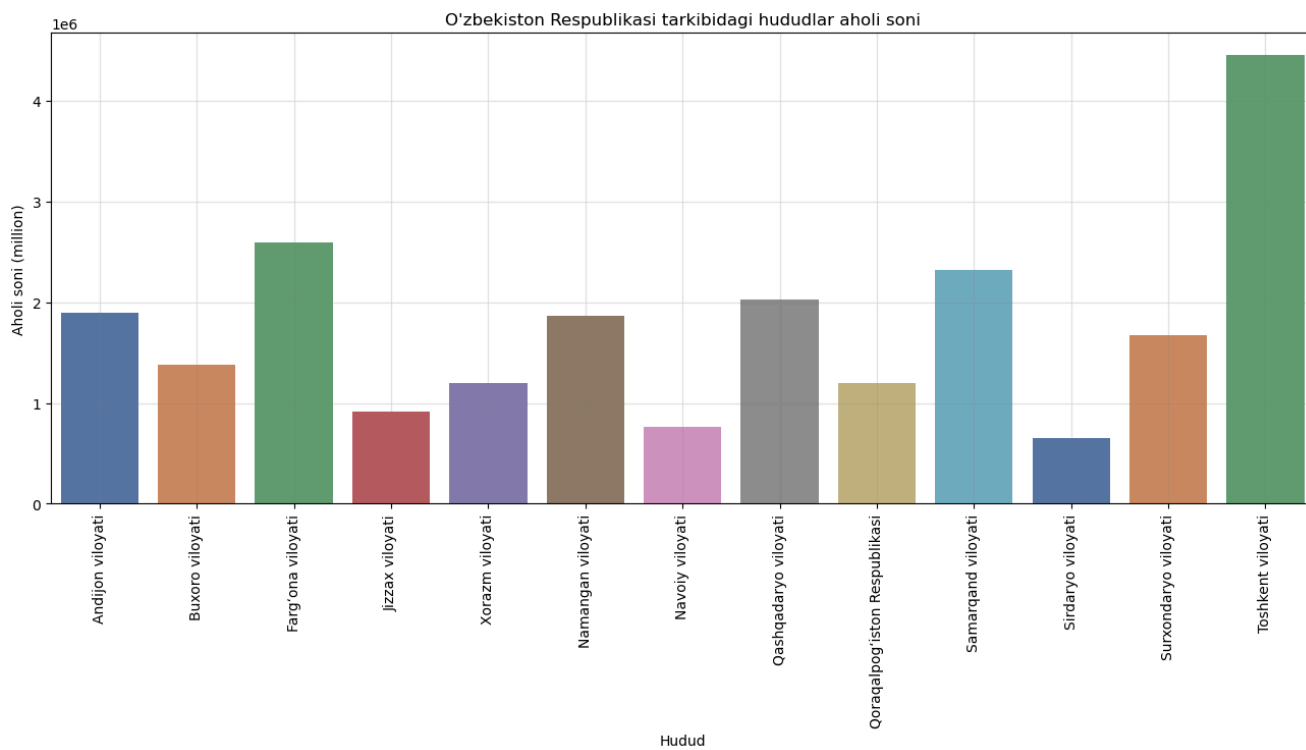


Barcha ustunlarni har xil rangda qilish

```
In [53]: plt.figure(figsize=(16, 6))

sns.barplot(
    x=df.index,
    y=df['Aholisi'],
    hue=df.index,
    palette=sns.color_palette("deep", len(df)),
    legend=False
)

plt.title("O'zbekiston Respublikasi tarkibidagi hududlar aholi soni")
plt.xlabel("Hudud")
plt.ylabel("Aholi soni (million)")
plt.grid(alpha=0.2, color="grey")
plt.xticks(rotation=90)
plt.show()
```



Barcha ustunlarni xohlagan ragnda to'ldirish

```
In [54]: plt.figure(figsize=(16, 6))

colors = ["red", "green", "blue", "orange", "purple", "brown",
          "pink", "gray", "yellow", "cyan", "magenta", "black", "lime"]

sns.barplot(
    x=df.index,
    y=df["Aholisi"],
    hue=df.index,
    palette=colors,
    legend=False
)

plt.title("O'zbekiston Respublikasi tarkibidagi hududlar aholi soni")
plt.xlabel("Hudud")
plt.ylabel("Aholi soni (million)")
plt.grid(alpha=0.2, color="grey")
plt.xticks(rotation=90)
plt.show()
```

